

JASHWANTH S

AI Engineer | Data Researcher | Data Engineer

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Professional Summary

Agentic AI Engineer and Data Scientist with hands-on experience building LLM-powered systems, autonomous agents, and scalable data pipelines. Proficient in Python, TensorFlow, LangChain, and cloud infrastructure with deep expertise in RAG architectures, LLM fine-tuning, NLP, and applied ML. Proven track record deploying end-to-end AI solutions — from intelligent agents to real-time inference systems — in enterprise environments.

Technical Skills

Languages: Python, SQL, R, JavaScript

LLMs & Agents: LangChain, HuggingFace Transformers, LLM Fine-tuning, RAG, AI Agents, Ollama, SentencePiece, FAISS, Prompt Engineering

ML / DL Frameworks: TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras

Computer Vision & NLP: OpenCV, CNN architectures, YOLO (v8/v10), Transfer Learning, Multi-model Fusion, NLTK, BLEU/ROUGE evaluation

MLOps & Cloud: Docker, AWS, GCP, SageMaker, FastAPI, Streamlit, Apache Airflow, Apache Spark

Data & Databases: Pandas, NumPy, MySQL, PostgreSQL, MongoDB, Elasticsearch

Specializations: Generative AI, Agentic Workflows, Predictive Modeling, Feature Engineering, Embedding Techniques, Model Evaluation

Work Experience

AI Engineer | Global Data

Aug 2025 – Present

- Built LLM-powered automated tagging workflows integrated with Spark, combining NER, TF-IDF/KeyBERT keyword extraction, and embedding-based semantic search to tag content at scale; used zero/few-shot LLM prompting via API for context-aware classification, reducing manual tagging effort by 70%.
- Automated data pipelines with Apache Airflow, orchestrating multi-step DAGs for ingestion, transformation, and validation across 1M+ rows weekly; incorporated retry logic, alerting, and dependency management to ensure fault-tolerant execution, cutting manual effort by 60%.
- Developed Spark-based enrichment pipelines handling 100GB+ datasets, applying distributed joins and custom UDFs to enrich records against internally validated mapping files and LLM-generated annotations, ensuring consistent, high-quality feature enrichment at scale.
- Built a data collection pipeline harvesting Google and YouTube search data using Playwright for browser automation with header caching and intelligent request throttling to avoid API rate limits; integrated LLM-based content classification and relevance scoring, with data stored in S3 and PostgreSQL, improving ingestion speed by 40%.
- Built and automated Classification and Prediction jobs for 10M+ data points across 200+ dashboards, incorporating metrics like CAGR, Rolling Windows, and MCP integrations, triggered monthly with intelligent monitoring and retry mechanisms.

Data Engineering Intern | Ai Palette

Aug 2024 – Apr 2025

- Managed 10M+ records across production datasets ensuring high availability; reduced data retrieval errors by 25%.
- Engineered optimized real-time processing workflows increasing system throughput by 35% and improving response times by 20%.
- Built scalable pipelines automating data movement to production, cutting manual processing time by 40% and enabling near real-time analytics.

Key Projects

Predictive Maintenance for IoT Devices

Jan 2025 – Mar 2025

- Engineered an ML system using Python, Scikit-learn, and TensorFlow to forecast IoT device failures with 92% accuracy and 87% recall across 500 simulated devices.
- Applied advanced feature extraction (rolling averages, FFT) and ensemble models (Random Forest, XGBoost, LSTM), achieving 15% improvement in recall over baseline.
- Deployed models via Flask on AWS with 1.2s average response time; built Streamlit dashboards processing 10,000+ data points per device.

Agentic RAG System

Oct 2023 – Dec 2023

- Designed and built a Retrieval-Augmented Generation (RAG) agent using Python, SentencePiece, and FAISS to process 5,000+ documents with 2.3s average retrieval time and 85% relevance accuracy.
- Integrated BLEU, ROUGE, and BLEURT evaluation metrics achieving BLEU 0.34 and ROUGE-L 0.42 across 500 test QA pairs — establishing a rigorous LLM output evaluation pipeline.
- Deployed end-to-end on Streamlit + HuggingFace Transformers supporting 10+ concurrent users with 95% uptime.

SpeakEasy: Real-Time Indian Language Translation

Jul 2024 – Dec 2024

- Fine-tuned transformer models on 100,000 parallel sentence pairs across 6 Indian languages, achieving 23.4 BLEU (English-Hindi) and 21.8 BLEU (English-Tamil).
- Delivered real-time translations at 1.8s latency per sentence with batch processing support via HuggingFace Transformers CLI and web interface.

Waste Management System – CNN Object Detection

Oct 2023 – Dec 2023

- Trained a CNN to classify 10 waste material categories with 92% accuracy on 15,000 images; augmented dataset 3× to improve cross-validation accuracy from 87% to 92%.
- Integrated with OpenCV for real-time inference at 150ms per image with 90% precision. Patent filed: 202341082049 A.

Education

BE (Honors) — Artificial Intelligence & Machine Learning

Visvesvaraya Technological University

Aug 2025

GPA: 9.53 CGPA | Ranked 10th in AIML Department | Award presented by Principal Secretary to the Government of Karnataka

GATE 2026 — DA Qualified

Score: 476 | Data Science and AI

Feb 2026

Certifications

- Explore Generative AI with the Gemini API in Vertex AI — Google Cloud
- Inspect Rich Documents with Gemini Multimodality & Multimodal RAG — Google Cloud
- Develop GenAI Apps with Gemini and Streamlit — Google Cloud
- Data Analytics Job Simulation — Quantum
- Data Science Job Simulation — BCG
- Machine Learning with Python — IBM